Dielectric tensor and its properties

$$\varepsilon_{ij}(\omega) = \varepsilon_0 \delta_{ij} + \frac{i}{\omega} \sigma_{ij}(\omega)$$

Joule's heating $P = \vec{j} \cdot \vec{E} \rightarrow j_i E_i$

$$\vec{j}(t) = \mathrm{Re}\left(\overleftrightarrow{\sigma}\, \vec{E}\right) \Rightarrow j_i(t) = \mathrm{Re}\left\{\sigma_{ij}(\omega) E_j e^{-i\omega t}\right\}$$

$$= \frac{1}{2}\left\{\sigma_{ij} E_j e^{-i\omega t} + \sigma_{ij}^* E_j^* e^{i\omega t}\right\}$$

$$\vec{E} \Rightarrow \mathrm{Re}(\vec{E}) = \frac{1}{2}\left\{E_i e^{-i\omega t} + E_i^* e^{i\omega t}\right\}$$

time averaged

$$\Rightarrow \overline{P} = \frac{1}{4}\left(\sigma_{ij} E_j E_i^* + \underbrace{\sigma_{ij}^* E_j^* E_i}_{\sigma_{ji}^* E_i^* E_j}\right) =$$

$$= \frac{1}{4}\left(\sigma_{ij} + \sigma_{ji}^*\right) E_j E_i^*$$

But $\sigma_{ij} = -i\omega\left(\varepsilon_{ij} - \varepsilon_0\right)$

$$\sigma_{ij} + \sigma_{ji}^* = -i\omega\left(\varepsilon_{ij} - \varepsilon_0\right) + i\omega\left(\varepsilon_{ji}^* - \varepsilon_0\right) =$$

$$= -i\omega\left(\varepsilon_{ij} - \varepsilon_{ji}^*\right)$$

$$\overline{P} = \frac{-i\omega}{4}\left(\varepsilon_{ij} - \varepsilon_{ji}^*\right) E_i^* E_j$$

Now we parameterize

$$\varepsilon_{ij}(\omega) = \frac{1}{2}\left(\varepsilon_{ij} + \varepsilon_{ji}^*\right) + i\,\frac{1}{2i}\left(\varepsilon_{ij} - \varepsilon_{ji}^*\right)$$

$$= a_{ij}(\omega) + i\, b_{ij}(\omega).$$

Consider $a_{ji}^* = \frac{1}{2}\left(\varepsilon_{ji}^* + \varepsilon_{ij}\right) = a_{ij}$

$$b_{ji}^* = \frac{1}{-2i}\left(\varepsilon_{ji}^* - \varepsilon_{ij}\right) = \frac{1}{2i}\left(\varepsilon_{ij} - \varepsilon_{ji}^*\right) = b_{ij}$$

So that $\varepsilon_{ji}^* = a_{ji}^* - i\, b_{ji}^* = a_{ij} - i\, b_{ij}$

(Note that $\varepsilon_{ij}^* = a_{ij}^* - i\, b_{ij}^* \neq \varepsilon_{ji}^*$)

Thus $\overline{P} = \frac{1}{2}\omega \frac{1}{2i}\left(\varepsilon_{ij} - \varepsilon_{ji}^*\right) E_i^* E_j = \frac{1}{2}\omega\, b_{ij}(\omega) E_i^* E_j$

Under time-reversal: $\left.\begin{array}{l} t \to -t \\ \omega \to -\omega \end{array}\right\}$ but $\overline{P}$ must remain positive

$$\Rightarrow \quad b_{ij}(\omega)\Big|_B = -b_{ij}(-\omega)\Big|_{-B}$$

Via Kramers-Kronig we find $a_{ij}(\omega)\Big|_B = a_{ij}(-\omega)\Big|_{-B}$

<u>Check</u>

$$\varepsilon_{ij}(\omega) = \varepsilon_0 \begin{pmatrix} 1 - \frac{\omega_p^2}{\omega^2} & i\,\frac{\omega_p^2 \Omega_c}{\omega^3} \\ -i\,\frac{\omega_p^2 \Omega_c}{\omega^3} & 1 - \frac{\omega_p^2}{\omega^2} \end{pmatrix}$$

Onsager relation

$$\boxed{\varepsilon_{ij}(\omega)\Big|_B = \varepsilon_{ji}^*(-\omega)\Big|_{-B} = \varepsilon_{ji}(\omega)\Big|_{-B}}$$

$$\varepsilon_{12}(\omega) = \frac{i\,\omega_p^2 \Omega_c}{\omega^3}$$

$$\varepsilon_{21}^*(\omega) = \left(-\frac{i\,\omega_p^2 \Omega_c}{\omega^3}\right)^* = \frac{i\,\omega_p^2 \Omega_c}{\omega^3}$$

$$\varepsilon_{21}^*(-\omega)\Big|_{-B} = \frac{i\,\omega_p^2(-\Omega_c)}{(-\omega)^3} = \frac{i\,\omega_p^2 \Omega_c}{\omega^3} = \varepsilon_{12}(\omega)$$

Kramers–Kronig

$$\varepsilon'(\Omega) - \varepsilon_0 = \frac{1}{\pi} P\int_{-\infty}^{\infty} d\omega \; \frac{\varepsilon''(\omega)}{\omega - \Omega}$$

$$a_{ij}(\Omega) - \varepsilon_0 \delta_{ij} = P\int \frac{d\omega}{\pi} \; \frac{b_{ij}(\omega)}{\omega - \Omega}$$

$$a_{ij}(-\Omega)\Big|_{-B} - \varepsilon_0 \delta_{ij} = P\int_{-\infty}^{\infty} \frac{d\omega}{\pi} \; \frac{b_{ij}(\omega)\big|_{-B}}{\omega + \Omega} = (\omega = -\omega') =$$

$$= P\int_{-\infty}^{\infty} \frac{d\omega'}{\pi} \; \frac{b_{ij}(-\omega')\big|_{-B}}{-\omega' + \Omega} =$$

$$= P\int \frac{d\omega}{\pi} \; \frac{b_{ij}(\omega)\big|_{B}}{\omega - \Omega} = a_{ij}(\Omega)\Big|_{B} - \varepsilon_0 \delta_{ij}$$

$\Rightarrow$ obtain that $a_{ij}(\omega)\big|_{B} = a_{ij}(-\omega)\big|_{-B}$

$$\varepsilon_{ij}(\omega)\big|_{B} = a_{ij}(\omega)\big|_{B} + i\, b_{ij}(\omega)\big|_{B} =$$

$$= a^*_{ji}(\omega)\big|_{B} + i\, b^*_{ji}(\omega)\big|_{B} \quad \{\text{hermicity}$$

$$= a^*_{ji}(-\omega)\big|_{-B} - i\, b^*_{ji}(-\omega)\big|_{-B} \quad \begin{cases}\text{oddness } \hat{b} \\ \text{evenness } \hat{a}\end{cases}$$

$$= \varepsilon^*_{ji}(-\omega)\big|_{-B} = \varepsilon_{ji}(\omega)\big|_{-B} \quad \{\text{reality}$$

Our Drude's form $\varepsilon_{ij}(\omega) = \varepsilon_0 n^2 \delta_{ij} + \varepsilon_0 i \dfrac{\omega_p^2 \Omega_c}{\omega^3} \varepsilon_{ijz}$

has this property as $\varepsilon_{jiz} = -\varepsilon_{ijz}$

and $\Omega_c\big|_{-B} = -\Omega_c\big|_{B}$.

Now let us check the combination:

$$\varepsilon_{ij} - \varepsilon_{ji}^{*} = 0 \text{ for } i=j$$

$$\varepsilon_{12} - \varepsilon_{21}^{*} = \frac{i\,\omega_p^2 \nu_c}{\omega^3} - \left(\frac{-i\,\omega_p^2 \nu_c}{\omega^3}\right)^{*} = 0$$

$$\varepsilon_{21} - \varepsilon_{12}^{*} = 0$$

$\Rightarrow$ there ~~are~~ is no joule's heating (energy loss)!

Why? $\Rightarrow$ we set $\tau = \infty$ from the very beginning!!!

With $\frac{1}{\tau} \neq 0$:

$$\frac{d\vec{v}}{dt} = \frac{q}{m}\vec{E}(\omega)e^{-i\omega t} + \frac{q}{m}\vec{v}\times\vec{B}_0 - \frac{\vec{v}}{\tau}$$

↳ static external field

$$\vec{v}\times\vec{B}_0 = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ v_x & v_y & v_z \\ 0 & 0 & B_0 \end{vmatrix} = \hat{x}\, v_y B_0 - \hat{y}\, v_x B_0$$

let us split $\vec{v} = \vec{v}_\perp + \vec{v}_z$

$$\vec{E} = \vec{E}_\perp + \vec{E}_z$$

And $\vec{v}(t) = \vec{v}(\omega)e^{-i\omega t}$

$$\left(-i\omega + \frac{1}{\tau}\right)\vec{v}_\perp = \frac{q}{m}\vec{E}_\perp(\omega) + \frac{q}{m}\vec{v}_\perp\times\vec{B}_0$$

$$\left(-i\omega + \frac{1}{\tau}\right)\vec{v}_\perp\times\vec{B}_0 = \frac{q}{m}\vec{E}_\perp\times\vec{B}_0 + \frac{q}{m}\left(\vec{v}_\perp\times\vec{B}_0\right)\times\vec{B}_0$$

$$-\vec{B}_0\times\left(\vec{v}_\perp\times\vec{B}_0\right)$$

$$= (-)\left(\vec{v}_\perp B_0^2 - \vec{B}_0\underbrace{\left(\vec{v}_\perp\cdot\vec{B}_0\right)}_{=0}\right)$$

$$\Rightarrow \left(-i\omega + \frac{1}{\tau}\right)\vec{v}_\perp\times\vec{B}_0 = \frac{q}{m}\vec{E}_\perp\times B_0 - \frac{qB_0^2}{m}\vec{v}_\perp$$

$$\Rightarrow \left(-i\omega + \frac{1}{\tau}\right)\vec{v}_\perp = \frac{q}{m}\vec{E}_\perp(\omega) + \frac{q}{m}\,\frac{\frac{q}{m}\vec{E}_\perp(\omega)\times\vec{B}_0 - \frac{q}{m}B_0^2\vec{v}_\perp}{-i\omega + \frac{1}{\tau}}$$

$$\left\{\left(-i\omega + \frac{1}{\tau}\right)^2 + \left(\frac{qB_0}{m}\right)^2\right\}\vec{v}_\perp(\omega) = \frac{q}{m}\left(-i\omega + \frac{1}{\tau}\right)\vec{E}_\perp + \frac{qB_0}{m}\vec{E}_\perp\times\hat{z}$$

For electrons $q = -e$, $\Omega_c = \frac{eB_0}{m}$

$$\vec{V}_\perp = -\frac{e}{m} \frac{\left(-i\omega + \frac{1}{\tau}\right)\vec{E}_\perp(\omega) + \Omega_c\, \hat{z} \times \vec{E}_\perp(\omega)}{\Omega_c^2 + \left(-i\omega + \frac{1}{\tau}\right)^2}$$

$$\hat{z} \times \vec{E}_\perp = \hat{z} \times \left(E_x \hat{x} + E_y \hat{y}\right) = E_x \hat{y} - E_y \hat{x}$$

$$\begin{pmatrix} V_x \\ V_y \end{pmatrix} = -\frac{e}{m} \frac{1}{\Omega_c^2 + \left(-i\omega + \frac{1}{\tau}\right)^2} \begin{pmatrix} -i\omega + \frac{1}{\tau} & -\Omega_c \\ \Omega_c & -i\omega + \frac{1}{\tau} \end{pmatrix} \begin{pmatrix} E_x \\ E_y \end{pmatrix}$$

$$\begin{pmatrix} j_x \\ j_y \end{pmatrix} = -en \begin{pmatrix} V_x \\ V_y \end{pmatrix} = \frac{e^2 n}{m} \frac{1}{\Omega_c^2 + \left(-i\omega + \frac{1}{\tau}\right)^2} \begin{pmatrix} -i\omega + \frac{1}{\tau} & -\Omega_c \\ \Omega_c & -i\omega + \frac{1}{\tau} \end{pmatrix} \begin{pmatrix} E_x \\ E_y \end{pmatrix}$$

$$\Rightarrow \quad j_i = \sigma_{ij} E_j$$

$$\sigma_{ij}^{\perp} = \sigma_{ij} = \frac{e^2 n}{m} \begin{pmatrix} \frac{-i\omega + \frac{1}{\tau}}{\Omega_c^2 + \left(-i\omega + \frac{1}{\tau}\right)^2} & \frac{-\Omega_c}{\Omega_c^2 + \left(-i\omega + \frac{1}{\tau}\right)^2} \\ \frac{\Omega_c}{\Omega_c^2 + \left(-i\omega + \frac{1}{\tau}\right)^2} & \frac{-i\omega + \frac{1}{\tau}}{\Omega_c^2 + \left(-i\omega + \frac{1}{\tau}\right)^2} \end{pmatrix} ; \quad \sigma = \begin{pmatrix} \sigma_{ij}^{\perp} & 0 \\ 0 & \sigma_{zz} \end{pmatrix}$$

3x3 matrix

whereas along z we only have $\sigma_{zz} = \frac{e^2 n}{m} \frac{1}{-i\omega + \frac{1}{\tau}}$

since Lorentz force does not affect V_z.

The Hall current: $\vec{j}$ perpendicular to $\vec{B}_0$ and $\vec{E}_\perp$

<u>Static limit</u> $\omega = 0$

$$\sigma^\perp(0) = \frac{e^2 n}{m} \frac{1}{\Omega_c^2 + \frac{1}{\tau^2}} \begin{pmatrix} \frac{1}{\tau} & -\Omega_c \\ \Omega_c & \frac{1}{\tau} \end{pmatrix} =$$

$$= \underbrace{\frac{e^2 n \tau}{m}}_{\sigma_0} \frac{1}{1+(\Omega_c \tau)^2} \begin{pmatrix} 1 & -\Omega_c \tau \\ \Omega_c \tau & 1 \end{pmatrix}$$

Hall conductivity $\sigma_H = \sigma_{xy} = \sigma_{12} = \sigma_0 \dfrac{-\Omega_c \tau}{1+(\Omega_c \tau)^2}$

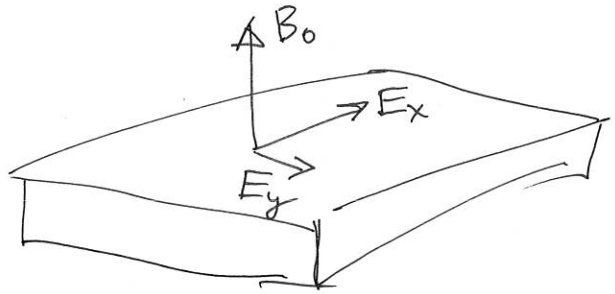
$$\xrightarrow[\Omega_c \tau \gg 1]{} -\frac{\sigma_0}{\Omega_c \tau} = -\frac{e^2 n \tau}{m} \frac{m}{e B_0 \tau} =$$

$$= -\frac{en}{B_0}$$

is independent of m, τ.

Depends on the sign of charge carriers: $(-e)$ for electrons,
$(+e)$ for holes.

Hall effect



Apply B_0 and E_x $\Rightarrow$ then current flows along $\hat{x}$ and $\hat{y}$ $\Rightarrow$ but current along $\hat{y}$ can not flow indefenitely as charges accumulate on the ~~sides~~ sides of the sample and result in E_y which stops the flow of j_y

$$\Rightarrow \quad j_y = 0 \quad , \text{but} \quad j_y = \sigma_{yx} E_x + \sigma_{yy} E_y$$

$$\Rightarrow \quad E_y = -\frac{\sigma_{yx}}{\sigma_{yy}} E_x$$

$$j_x = \sigma_{xx} E_x + \sigma_{xy} E_y = \left(\sigma_{xx} - \frac{\sigma_{xy}\sigma_{yx}}{\sigma_{yy}}\right) E_x$$

$$= \frac{E_x}{\rho_{xx}}$$

Resistivity tensor $\rho = \sigma^{-1}$

$$\begin{pmatrix} \rho_{xx} & \rho_{xy} \\ \rho_{yx} & \rho_{yy} \end{pmatrix} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{pmatrix}^{-1} = \frac{1}{\sigma_{xx}\sigma_{yy} - \sigma_{xy}\sigma_{yx}} \begin{pmatrix} \sigma_{yy} & -\sigma_{xy} \\ -\sigma_{yx} & \sigma_{xx} \end{pmatrix}$$

$$\Rightarrow \rho_{xx} = \frac{\sigma_{yy}}{\sigma_{xx}\sigma_{yy} - \sigma_{xy}\sigma_{yx}} = \frac{1}{\sigma_{xx} - \frac{\sigma_{xy}\sigma_{yx}}{\sigma_{yy}}}$$

$$\sigma(0) = \frac{\sigma_0}{1+(\Omega_c\tau)^2}\begin{pmatrix} 1 & -\Omega_c\tau \\ \Omega_c\tau & 1 \end{pmatrix} \quad , \quad \sigma_{xy} = -\sigma_{yx}$$

$$\sigma_{xx} = \sigma_{yy}$$

Hence $\sigma_{xx} - \dfrac{\sigma_{xy}\sigma_{yx}}{\sigma_{yy}} = \sigma_{xx} + \dfrac{\sigma_{xy}^2}{\sigma_{xx}} = \dfrac{\sigma_0}{1+(\Omega_c\tau)^2} + \dfrac{\left(\dfrac{\sigma_0\Omega_c\tau}{1+(\Omega_c\tau)^2}\right)^2}{\dfrac{\sigma_0}{1+(\Omega_c\tau)^2}}$

$$= \frac{\sigma_0}{1+(\Omega_c\tau)^2} + \frac{\sigma_0(\Omega_c\tau)^2}{1+(\Omega_c\tau)^2} = \sigma_0$$

$$\Rightarrow \rho_{xx} = \rho_{yy} = \frac{1}{\sigma_0}$$

$$\rho = \rho_0\begin{pmatrix} 1 & (\Omega_c\tau)^{-1} \\ -(\Omega_c\tau)^{-1} & 1 \end{pmatrix} \quad , \rho_0 = \frac{1}{\sigma_0}$$

$\Rightarrow j_x = \sigma_0 E_x$ is not effected by B_0 !

Hall effect: $R_H = \dfrac{V_y}{I_x} = \dfrac{E_y L_y}{j_x L_y L_z} = \dfrac{E_y}{j_x L_z} = -\dfrac{E_x \sigma_{yx}}{j_x \sigma_{yy} L_z}$

$$= -\frac{E_x \sigma_{yx}}{\sigma_0 E_x \sigma_{yy} L_z} = -\frac{\sigma_{yx}}{\sigma_0 \sigma_{yy} L_z} = \frac{-\Omega_c\tau}{\sigma_0 L_z} =$$

$$= \frac{-\frac{eB_0}{m}\tau}{\frac{e^2 n\tau}{m} L_z} = -\frac{B_0}{enL_z}$$

Cyclotron resonance: AC field, $\omega \neq 0$

Assume clean conductor so that $\omega \gg \frac{1}{\tau}$

Suppose also $\vec{E}_\perp = \hat{x}\, E_0\, e^{-i\omega t}$ (linearly polarized external field)

$$\Rightarrow \overline{P} = \frac{1}{4}\left(\sigma_{ij} + \sigma_{ji}^*\right) E_i^* E_j \rightarrow \frac{1}{2}\sigma'_{xx}\,|E_0|^2$$

$$\sigma'_{xx} = \frac{e^2 n}{m}\, \mathrm{Re}\left(\frac{-i\omega + \frac{1}{\tau}}{\Omega_c^2 + \left(-i\omega + \frac{1}{\tau}\right)^2}\right) = \frac{e^2 n \tau}{m} \cdot$$

$$\cdot\, \mathrm{Re}\left(\frac{1 - i\omega\tau}{(\Omega_c\tau)^2 + (1 - i\omega\tau)^2}\right)$$

$$\mathrm{Re}\,\frac{1 - i\omega\tau}{(\Omega_c\tau)^2 + 1 - (\omega\tau)^2 - i\,2\omega\tau} =$$

$$= \mathrm{Re}\,\frac{(1 - i\omega\tau)\left(1 + (\Omega_c^2 - \omega^2)\tau^2 + i\,2\omega\tau\right)}{\left(1 + (\Omega_c^2 - \omega^2)\tau^2\right)^2 + 4\omega^2\tau^2}$$

$$= \frac{1 + (\Omega_c^2 - \omega^2)\tau^2 + 2\omega^2\tau^2}{1 + (\Omega_c^2 - \omega^2)^2\tau^4 + 2(\Omega_c^2 - \omega^2)\tau^2 + 4\omega^2\tau^2}$$

$$= \frac{1 + (\Omega_c^2 + \omega^2)\tau^2}{1 + (\Omega_c^2 - \omega^2)^2\tau^4 + 2(\Omega_c^2 + \omega^2)\tau^2}$$

Can neglect 1 relative to $\omega\tau \gg 1$, $\Omega_c\tau \gg 1$

$$\Rightarrow \sigma'_{xx} = \sigma_0 \frac{(\Omega_c^2+\omega^2)\tau^2}{(\Omega_c^2-\omega^2)^2\tau^4 + 2(\Omega_c^2+\omega^2)\tau^2}$$

Note: for $\tau \to \infty$ $\sigma'_{xx} \to \sigma_0 \dfrac{(\Omega_c^2+\omega^2)}{(\Omega_c^2-\omega^2)^2\tau^2} \to 0$

so that absorption is determined by τ.

But this result also has another nice feature $\longrightarrow$ resonance:

for $\Omega_c \approx \omega$ $\quad \sigma'_{xx} = \sigma_0 \dfrac{2\omega^2\tau^2}{4\omega^2\tau^2(\Omega_c-\omega)^2\tau^2 + 4\omega^2\tau^2}$

$\Omega_c^2-\omega^2 = (\Omega_c-\omega)(\Omega_c+\omega)$

$\cong 2\omega(\Omega_c-\omega)$

$$= \frac{1}{2}\sigma_0 \frac{1}{1+(\Omega_c-\omega)^2\tau^2}$$

$$\Rightarrow \overline{P}(\omega) \Rightarrow \frac{1}{4}\frac{\sigma_0}{1+(\omega-\Omega_c)^2\tau^2}|E_0|^2$$

Absorption has maximum at $\omega = \Omega_c$ and is reduced by $\frac{1}{2}$ at $\omega = \Omega_c \pm \frac{1}{\tau}$.

Also: the resonance is <u>chiral</u>

$$\vec{E}_\perp = \hat{x}\, E_0 e^{-i\omega t} \pm i\,\hat{y}\, E_0 e^{-i\omega t}$$

$$\overline{P} = \frac{1}{4}\left(2\sigma'_{xx} + 2\sigma'_{yy}\right)|E_0|^2 + \frac{1}{4}\left(\sigma_{xy} + \sigma^*_{yx}\right) E_0^* \cdot (\pm i E_0)$$

$$+ \frac{1}{4}\left(\sigma_{yx} + \sigma^*_{xy}\right)(\pm i E_0)^* E_0 = \left(\text{use } \sigma_{yx} = -\sigma_{xy}\right)$$

$$= \sigma'_{xx}|E_0|^2 + \frac{1}{4}\left(\sigma_{xy} - \sigma^*_{xy}\right)(\pm i)|E_0|^2$$

$$+ \frac{1}{4}\left(-\sigma_{xy} + \sigma^*_{xy}\right)(\mp i)|E_0|^2 =$$

$$= \left[\sigma'_{xx} + \frac{1}{4}\left(2i\,\sigma''_{xy}(\pm i) - \frac{1}{4}\,2i\,\sigma''_{xy}(\mp i)\right)\right]|E_0|^2$$

$$= \left(\sigma'_{xx} \mp \sigma''_{xy}\right)|E_0|^2.$$

$$\sigma''_{xy} = \mathrm{Im}\,\sigma_0\left(\frac{-\Omega_c\tau}{(\Omega_c\tau)^2 + (1-i\omega\tau)^2}\right) = \sigma_0(-\Omega_c\tau)\cdot$$

$$\cdot\,\mathrm{Im}\left[\frac{1 + (\Omega_c^2 - \omega^2)\tau^2 + i\,2\omega\tau}{\left(1 + (\Omega_c^2-\omega^2)\tau^2\right)^2 + 4\omega^2\tau^2}\right] = \frac{-\sigma_0\,\Omega_c\,\tau\,(2\omega\tau)}{(\Omega_c^2-\omega^2)^2\tau^4 + 2(\Omega_c^2+\omega^2)\tau^2}$$

$$\Rightarrow \sigma'_{xx} \mp \sigma''_{xy} \sim (\Omega_c^2 + \omega^2)\tau^2 \mp (-2\,\Omega_c\,\omega\,\tau^2)$$

$$= (\Omega_c \cdot \pm\omega)^2\tau^2$$

So that $\overline{P} = \sigma_0 \dfrac{(\Omega_c \pm \omega)^2\tau^2}{4\omega^2\tau^2\left[1 + (\Omega_c - \omega)^2\tau^2\right]}|E_0|^2$

for $\omega \approx \Omega_c$

$$= \frac{1}{4}\sigma_0|E_0|^2\,\frac{\left(1 \pm \frac{\Omega_c}{\omega}\right)^2}{1 + (\Omega_c - \omega)^2\tau^2}$$

The resonance is <u>chiral</u>:
enhanced for $\oplus$ circular polarization and suppressed for $\ominus$.